# Helena Mannochio Russo

Postdoctoral Researcher · Skaggs School of Pharmacy and Pharmaceutical Sciences, University of California San Diego

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## RESEARCH INTERESTS

My primary research interests center on LC-MS/MS-based untargeted metabolomics, computational tools, big data mining, and artificial intelligence (AI), with a strong emphasis on developing innovative analytical approaches to understand complex biological systems. During my doctoral research, I applied LC-MS/MS-based untargeted metabolomics to investigate plant metabolism, integrating chemotaxonomic analysis with natural product discovery to uncover unusual metabolites with bioactive roles. As a postdoctoral researcher in the Dorrestein Lab, I expanded my expertise into computational metabolomics, developing large-scale spectral libraries, mining public data repositories, and creating community-accessible resources to investigate both microbial and plant-derived metabolites. I also develop AI-driven approaches for metabolomics, including Brain-to-AI, a project that captures the expert knowledge used to elucidate chemical structures from MS/MS spectra. In my independent research program, I aim to integrate computational and experimental approaches to discover new molecules and elucidate their biological functions, with the overarching goal of defining how they shape human health.

**Keywords:** Untargeted LC-MS/MS metabolomics, data mining, artificial intelligence, machine learning, database development, structural elucidation, microbial metabolites, plant metabolites, multi-omics, natural products, host signaling, human health.

## EDUCATION / RESEARCH APPOINTMENTS

**Postdoctoral Researcher** August 2022 – Present
Mentor: Dr. Pieter Dorrestein
University of California San Diego (San Diego, CA, United States of America)

**Doctor of Philosophy in Chemistry** March 2018 – April 2022
Advisor: Dr. Vanderlan da Silva Bolzani
Institute of Chemistry, São Paulo State University (Araraquara, SP, Brazil)

**Master's in Chemistry** August 2015 – March 2018
Advisors: Dr. Vanderlan da Silva Bolzani and Dr. Emerson Ferreira Queiroz
Institute of Chemistry, São Paulo State University (Araraquara, SP, Brazil)

**Bachelor in Chemistry** March 2010 – August 2015
Advisors: Dr. Julio Cezar Pastre and Dr. Ronaldo Aloise Pilli
Institute of Chemistry, University of Campinas (Campinas, SP, Brazil)

## GRANTS

**NIH K99/R00 Pathway to Independence Award** 2026
National Institute of Neurological Disorders and Stroke (NINDS), NIH. Principal Investigator.
Grant: 1K99NS147878-01A1
*Impact of microbial metabolites and their roles in HIV-associated neurocognitive disorders*

**Center Pilot Project** 2026

Polysubstance Addiction Research Center (PARC), supported by the National Institute on Drug Abuse (NIDA), NIH. Principal Investigator.

*Gut-brain metabolic signatures of oxycodone exposure*

## SKILLS

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|---|---|
| **Computational** | Big data mining; development of web applications for metabolomics research; database development; MS/MS data acquisition, processing, and analysis; Python; R; Bash. |
| **Wet lab** | Sample preparation for LC-MS-based metabolomics; chromatographic method development, optimization, and validation; fractionation and purification of natural products; structure elucidation; organic synthesis. |
| **Software** | Bruker Data Analysis; Thermo Xcalibur; GNPS; QIIME2; Illustrator; MZmine; Bruker Topspin; ChemDraw; Cytoscape. |
| **Soft skills** | Teamwork; leadership; interpersonal skills; scientific communication. |
| **Languages** | Portuguese (native); English (fluent); French (basic-intermediate); Spanish (basic). |

## PROJECTS

**Computational Metabolomics and Resource Development** 2022 – Present

Postdoctoral project; Dr. Pieter C. Dorrestein. Skaggs School of Pharmacy and Pharmaceutical Sciences, University of California San Diego

- **Development of spectral libraries from public metabolomics repositories:** I created comprehensive MS/MS spectral libraries by searching fragmentation patterns in public metabolomics datasets. I applied this computational approach to three classes of metabolites, and this allowed the creation of spectral libraries containing >5,500 putative bile acids, 851 *N*-acyl lipids, and >2,500 carnitines, representing the largest publicly available resources for metabolomic annotation of these compound classes. These libraries guided targeted synthesis and structural confirmation of dozens of previously unreported molecules, significantly expanding knowledge of metabolic diversity within these chemical classes.
- **Microbial metabolite knowledgebase development:** I lead the development of the Collaborative Microbial Metabolite Center knowledgebase (CMMC-kb), a resource that integrates spectral and structural data of microbial metabolites with biological information, including microbial producers, bioactivity, and biosynthesis. This comprehensive resource accelerates metabolite annotation and contextualization while enabling mechanistic investigation of microbiome function, advancing both molecular identification and biological interpretation of metabolomics results.
- **Plant metabolomics database development:** I developed plantMASST, a reference database that integrates 41,727 taxonomically defined plant extracts from 3,568 species. This database provides a foundation for large-scale chemotaxonomic investigations, allowing researchers to trace metabolite distributions across plant species using public metabolomics data. plantMASST has already enabled the discovery of previously uncharacterized plant-derived peptides, demonstrating its value for advancing natural products research.
- **Development of open science tools:** I developed MetaboApps, a suite of modular web applications integrated into the GNPS ecosystem. These web applications are designed to make complex data analysis in metabolomics more accessible to the scientific community, as they can be used not only by beginners or experienced data scientists, but also clinicians, microbiologists, among others.
- **AI for structural elucidation (Brain-to-AI):** I lead Brain-to-AI, a project that captures the expert knowledge used to elucidate chemical structures from MS/MS spectra. Much of this reasoning is never formally documented, and instead lives in laboratory notebooks, in annotated spectra, and in the explanations experts give when asked how they reached a conclusion. We capture it from handwritten notes, annotated spectra, and recorded interviews with specialists, and turn it into a skill that enables an AI assistant to annotate molecules from MS/MS data, grade its confidence, and cite the source of every inference.

**Chemical profiling of plant species for chemotaxonomic investigations** 2018 – 2022

Doctoral thesis project; Dr. Vanderlan da Silva Bolzani. Institute of Chemistry, São Paulo State University

I conducted a chemotaxonomic investigation of the Malpighiaceae botanical family using computational metabolomics tools to correlate classes of compounds with evolutionary patterns. During this work, I also applied LC-MS/MS to guide the isolation of unusual nitropropanoyl glucosides from plants, which showed larvicidal activity against *Aedes aegypti* larvae.

**Phytochemical investigation of *Niedenzuella multiglandulosa*** 2015 – 2018

Master's thesis project; Dr. Vanderlan da Silva Bolzani and Dr. Emerson Ferreira Queiroz. Institute of Chemistry, São Paulo State University

I performed the first phytochemical investigation of *Niedenzuella multiglandulosa*, an endemic Brazilian plant toxic to cattle. My research involved isolating and structurally characterizing the natural products accumulated in the leaves using NMR and LC-MS/MS.

## FELLOWSHIPS / SCHOLARSHIPS

**PhD research project scholarship** 2018 – 2022
National Council for Scientific and Technological Development (CNPq)

**Master's research project scholarship** 2015 – 2017
National Council for Scientific and Technological Development (CNPq)

**Scholarship for undergraduate research project** 2014 – 2015
Education, Research, and Extension Support Fund (FAEPEX)

## HONORS / AWARDS

**Finalist, Dream Biology Award** 2026
Organized by the Institute of Organic Chemistry and Biochemistry of the Czech Academy of Sciences (IOCB Prague)

**Interdisciplinary Research Fellowship in NeuroAIDS (IRFN)** 2026
Sponsored by the HIV Neurobehavioral Research Center. *Awarded; declined upon receipt of the NIH K99/R00 Pathway to Independence Award, as concurrent training support is not permitted.*

**Rising Star in Computational Mass Spectrometry** 2025
Sponsored by the Chan Zuckerberg Initiative, OpenMS, and Females in Mass Spectrometry

**MANA ECM Travel Award for the MANA 2025 conference** 2025
Sponsored by the Metabolomics Association of North America

**Poster award, 2025 Gordon Research Conference - Metabolomics and Human Health** 2025
Ventura, CA, USA

**MANA ECM Travel Award for the 2024 Metabolomics Society Conference** 2024
Sponsored by the Metabolomics Association of North America

**Selected for the 71st Lindau Nobel Laureate Meeting (dedicated to Chemistry)** 2022
Brazilian Academy of Sciences

**Selected for the Baden-Württemberg Post Conference Program of the Lindau Nobel Laureate Meeting** 2022
Baden-Württemberg International

**CAS Future Leaders in Chemistry** 2022
Sponsored by the American Chemical Society

**UNESP Women in Science Award, graduate student category – Exact and Earth Sciences** 2022
Institute of Chemistry, São Paulo State University (UNESP)

**Fulbright Doctoral Dissertation Research Award (DDRA)** 2019 – 2020
Sponsored by the Fulbright Commission

**Poster award, Natural Products Chemistry Division** 2019
42nd Annual Meeting of the Brazilian Chemical Society

**Poster award** 2017
VIII Symposium of Thermal Analysis

## PUBLICATIONS

62 published articles, 17 as first or co-first author. † co-first authors · * corresponding author

62. **Fecal Metabolomics-Based Dietary Profiling Reveals Food Markers of HIV Brain Health Disorders**
Crystal X. Wang, Harsha Gouda, Toon A.W. Scheurink, Jasmine Zemlin, Christine E. Spaven, **Helena Mannochio-Russo**, Jennifer E. Iudicello, Robert K. Heaton, David J. Moore, Scott L. Letendre, Mariana Cherner, Donald R. Franklin, Monica Guma, Andrew H. Miller, Rob Knight, Pieter C. Dorrestein, Ronald J. Ellis
*The Journal of Nutrition*, 2026. doi.org/10.1016/j.tjnut.2026.101873

61. **Synthesis-driven reverse metabolomics reveals 3-hydroxy *N*-acyl amides as gut microbial molecules**
Victoria Deleray, Vincent Charron-Lamoureux, Kyle Vittali, **Helena Mannochio-Russo**, Crystal X. Wang, Corinn Walker, Karsten Zengler, Dionicio Siegel, Pieter C. Dorrestein*
*mSystems*, 2026. doi.org/10.1128/msystems.00839-26

60. **Gut microbiome and metabolite profiles are associated with subsequent histologic MASH in a longitudinal mouse model**
Stephany Flores Ramos, Kelly Fogelson, Valentina B. Muti, Allegra T. Aron, Rodolfo A. Salido, R. Alexander Richter, **Helena Mannochio-Russo**, Jingjing Hu, Mojgan Hosseini, Rohit Loomba, Pieter C. Dorrestein, Rob Knight, Amir Zarrinpar*
*mSystems*, 2026. doi.org/10.1128/msystems.00720-26

59. **Bridging complexity and accessibility in metabolomics with MetaboApps**
**Helena Mannochio-Russo**†, Wilhan D. Gonçalves Nunes†, Haoqi Nina Zhao, Kine Eide Kvitne, Shipei Xing, Harsha Gouda, Julius Agongo, Ipsita Mohanty, Vincent Charron-Lamoureux, Prajit Rajkumar, Abzer K. Pakkir Shah, Axel Walter, Rithi Krishnaraj, Yasin El Abiead, Patrick C. Ferreira, Simone Zuffa, Abubaker Patan, Andrés Mauricio Caraballo-Rodríguez, Wout Bittremieux, Daniel Petras, Mingxun Wang*, Pieter C. Dorrestein*
*Nature Methods*, Vol. 23, No. 8, 2026. doi.org/10.1038/s41592-026-03186-8

58. **Potential role of bile acids as a microbiome-derived mechanism in synovitis of knee osteoarthritis synovitis**
Jessica D. Murillo-Saich†, **Helena Mannochio-Russo**†, Marta Sala-Climent, Nathan Argel, Anna Quan, Michal Kalli Hose, Rocio Paz-Gonzalez, Meghana Akkati, Eric Chang, Adnan Cutuk, Emily Gentry, Roxana Coras, Nancy E. Lane, Pieter C. Dorrestein, Monica Guma*
*Osteoarthritis and Cartilage*, Vol. 34, No. 6, 2026. doi.org/10.1016/j.joca.2026.02.011

57. **Glucuronidation metabolomic fingerprinting to map host-microbe metabolism**
Nina R. Boyle, Josh J. Sekela, Mingxun Wang, **Helena Mannochio-Russo**, Jeong Joo Pyo, Min Soo Kim, Shuchang Tian, Imhoi Koo, Elliot S. Friedman, Ceylan Tanes, Mallappa Anitha, Yuan Tian, Ethan W. Morgan, Iain A. Murray, Joseph P. Zackular, Kyle Bittinger, James D. Lewis, Gary H. Perdew, Gary D. Wu, Babette S. Zemel, Pieter C. Dorrestein, Jordan E. Bisanz, Matthew R. Redinbo, Andrew D. Patterson*
*Nature Communications*, Vol. 17, No. 1, 2026. doi.org/10.1038/s41467-026-73398-1

56. **Structure-centric searching enables global mapping of the public metabolome**
Yasin El Abiead*, Jeong In Seo, Vincent Charron-Lamoureux, Michael Strobel, Wilhan Donizete Gonçalves Nunes, Haoqi Nina Zhao, Kine Eide Kvitne, Simone Zuffa, **Helena Mannochio-Russo**, Harsha Gouda, Cristina Bez, Abubaker Patan, Shipei Xing, Jasmine Zemlin, Ipsita Mohany, Julius Agongo, Andres Mauricio Caraballo Rodriguez, Lindsey A. Burnett, Victoria Deleray, Abzer K. Pakkir Shah, Jarmo-Charles Kalinski, Daniel Petras, Nikiforos Alygizakis, Jeremy Carver, Ozgur Yurekten, Thomas Payne, Eoin Fahy, Shankar Subramaniam, Juan Antonio Vizcaíno, Mingxun Wang*, Pieter C. Dorrestein*
*Nature Biotechnology*, 2026. doi.org/10.1038/s41587-026-03082-8

55. **Systemic rhythmicity of host and bacterial bile acid amidates in the mouse**
Erin R. Reilly, Vincent Charron-Lamoureux, **Helena Mannochio-Russo**, Ethan W. Morgan, Nina R. Boyle, Imhoi Koo, Emma Coudriet, Fuhua Hao, Ipsita Mohanty, Gary H. Perdew, Pieter C. Dorrestein, Andrew D. Patterson*
*Cell Systems*, Vol. 17, No. 4, 2026. doi.org/10.1016/j.cels.2026.101541

54. **MS/MS Mass Spectrometry Filtering Tree for Bile Acid Regio- and Stereoisomer Annotation**
Ipsita Mohanty, Shipei Xing, Vanessa Castillo, Julius Agongo, Abubaker Patan, Yasin El Abiead, **Helena Mannochio-Russo**, Wilhan D. Gonçalves Nunes, Jasmine Zemlin, Itzhak Mizrahi, Dionicio Siegel, Mingxun Wang, Lee R. Hagey, Pieter C. Dorrestein*
*Analytical Chemistry*, Vol. 98, No. 6, 2026. doi.org/10.1021/acs.analchem.5c05677

53. **Antiglycation Activity and LC-MS/MS-Based Metabolomic Annotation of *Psidium cattleianum* (Myrtaceae) Fruit Peels**
Anna Beatriz Sabino Ferrari, Giselle Lopes Silva, **Helena Mannochio-Russo**, Jaqueline Munise Guimarães da Silva, Martina Herrmann, Rodrigo Sant'ana Cabral, Maria Luiza Zeraik*
*ACS Food Science & Technology*, Vol. 6, No. 2, 2026. doi.org/10.1021/acsfoodscitech.5c00842

52. **Emerging Microbial Lipid Shifts in a Multistage Simulated Gut Inoculated with Human Fecal Microbiota**
Ifrat Tamanna, Katja Salonen, **Helena Mannochio-Russo**, Matilda Kråkström, Paulo Wender P. Gomes, Vincent Charron-Lamoureux, Ipsita Mohanty, Sofia D. Forssten, Arthur C. Ouwehand, Tuulia Hyötyläinen, Alex M. Dickens, Pieter C. Dorrestein, Matej Orešič, Santosh Lamichhane*
*Journal of Proteome Research*, Vol. 25, No. 9, 2026. doi.org/10.1021/acs.jproteome.6c00050

51. **Fecal microbial and metabolic signatures in children with very early onset inflammatory bowel disease**
Kine Eide Kvitne, Simone Zuffa, Vincent Charron-Lamoureux, Ipsita Mohanty, Abubaker Patan, **Helena Mannochio-Russo**, Jasmine Zemlin, Lindsey A. Burnett, Lisa S. Zhang, Mia C. Cecala, Ceylan Ersoz, James A. Connelly, Natasha Halasa, Maribeth Nicholson, Pieter C. Dorrestein, Shirley M. Tsunoda, Janet Markle*
*npj Biofilms and Microbiomes*, Vol. 12, No. 1, 2025. doi.org/10.1038/s41522-025-00899-0

50. **A resource to empirically establish drug exposure records directly from untargeted metabolomics data**
Haoqi Nina Zhao, Kine Eide Kvitne, Corinna Brungs, Siddharth Mohan, Vincent Charron-Lamoureux, Wout Bittremieux, Runbang Tang, Robin Schmid, Santosh Lamichhane, Shipei Xing, Yasin El Abiead, Mohammadsobhan S. Andalibi, **Helena Mannochio-Russo**, Madison Ambre, Nicole E. Avalon, MacKenzie Bryant, Lindsey A. Burnett, Andrés Mauricio Caraballo-Rodríguez, Martin Casas Maya, Loryn Chin, Lluís Corominas, Ronald J. Ellis, Donald Franklin, Sagan Girod, Paulo Wender P. Gomes, Lauren Hansen, Robert K. Heaton, Jennifer E. Iudicello, Alan K. Jarmusch, Lora Khatib, Scott Letendre, Sarolt Magyari, Daniel McDonald, Ipsita Mohanty, Andrés Cumsille, David J. Moore, Prajit Rajkumar, Dylan H. Ross, Harshada Sapre, Mohammad Reza Zare Shahneh, Ruben Gil-Solsona, Sydney P. Thomas, Caitlin Tribelhorn, Helena M. Tubb, Corinn Walker, Crystal X. Wang, Jasmine Zemlin, Simone Zuffa, David S. Wishart, Pablo Gago-Ferrero, Rima Kaddurah-Daouk, Mingxun Wang, Manuela Raffatellu, Karsten Zengler, Tomáš Pluskal, Libin Xu, Rob Knight, Shirley M. Tsunoda, Pieter C. Dorrestein*
*Nature Communications*, Vol. 16, No. 1, 2025. doi.org/10.1038/s41467-025-65993-5

49. **Targeted Microbial Shifts and Metabolite Profiles Were Associated with Clinical Response to an Anti-Inflammatory Diet in Osteoarthritis**
Marta Sala-Climent, Kevin Bu, Roxana Coras, Martha Cedeno, Simone Zuffa, Jessica Murillo-Saich, **Helena Mannochio-Russo**, Celeste Allaband, Michal K. Hose, Anna Quan, Soo-In Choi, Katherine Nguyen, Shahrokh Golshan, Rebecca B. Blank, Tiffany Holt, Nancy E. Lane, Rob Knight, Jose Scher, Pieter Dorrestein, Jose Clemente*, Monica Guma*
*Nutrients*, Vol. 17, No. 17, 2025. doi.org/10.3390/nu17172729

48. **The microbiome diversifies long- to short-chain fatty acid-derived N-acyl lipids**
**Helena Mannochio-Russo**, Vincent Charron-Lamoureux, Martijn van Faassen, Santosh Lamichhane, Wilhan D. Gonçalves Nunes, Victoria Deleray, Adriana V. Ayala, Yuichiro Tanaka, Abubaker Patan, Kyle Vittali, Prajit Rajkumar, Yasin El Abiead, Haoqi Nina Zhao, Paulo Wender Portal Gomes, Ipsita Mohanty, Carlynda Lee, Aidan Sund, Meera Sharma, Yuanhao Liu, David Pattynama, Gregory T. Walker, Grant J. Norton, Lora Khatib, Mohammadsobhan S. Andalibi, Crystal X. Wang, Ronald J. Ellis, David J. Moore, Jennifer E. Iudicello, Donald Franklin, Scott Letendre, Loryn Chin, Corinn Walker, Simone Renwick, Jasmine Zemlin, Michael J. Meehan, Xinyang Song, Dennis Kasper, Zachary Burcham, Jane J. Kim, Sejal Kadakia, Manuela Raffatellu, Lars Bode, Hiutung Chu, Karsten Zengler, Mingxun Wang, Dionicio Siegel, Rob Knight, Pieter C. Dorrestein*
*Cell*, Vol. 188, No. 15, 2025. doi.org/10.1016/j.cell.2025.05.015

47. **Enabling pan-repository reanalysis for big data science of public metabolomics data**
Yasin El Abiead, Michael Strobel, Thomas Payne, Eoin Fahy, Claire O'Donovan, Shankar Subramamiam, Juan Antonio Vizcaíno, Ozgur Yurekten, Victoria Deleray, Simone Zuffa, Shipei Xing, **Helena Mannochio-Russo**, Ipsita Mohanty, Haoqi Nina Zhao, Andres M. Caraballo-Rodriguez, Paulo Wender P. Gomes, Nicole E. Avalon, Trent R. Northen, Benjamin P. Bowen, Katherine B. Louie, Pieter C. Dorrestein, Mingxun Wang*
*Nature Communications*, Vol. 16, No. 1, 2025. doi.org/10.1038/s41467-025-60067-y

46. ***Ex Vivo* Interaction Between Human Gut Microbiota and Artemisinin: A Multi-Omics Perspective**
Paulo Wender Portal Gomes*, **Helena Mannochio-Russo**, Stanislav N. Iablokov, Dmitry A. Rodionov, Heejung Yang, Pieter C. Dorrestein, Scott N. Peterson, Christine Tara Peterson*
*ACS Omega*, Vol. 10, No. 21, 2025. doi.org/10.1021/acsomega.5c01983

45. **A universal language for finding mass spectrometry data patterns**
Tito Damiani, Alan K. Jarmusch, Allegra T. Aron, Daniel Petras, Vanessa V. Phelan, Haoqi Nina Zhao, Wout Bittremieux, Deepa D. Acharya, Mohammed M. A. Ahmed, Anelize Bauermeister, Matthew J. Bertin, Paul D. Boudreau, Ricardo M. Borges, Benjamin P. Bowen, Christopher J. Brown, Fernanda O. Chagas, Kenneth D. Clevenger, Mario S. P. Correia, William J. Crandall, Max Crüsemann, Eoin Fahy, Oliver Fiehn, Neha Garg, William H. Gerwick, Jeffrey R. Gilbert, Daniel Globisch, Paulo Wender P. Gomes, Steffen Heuckeroth, C. Andrew James, Scott A. Jarmusch, Sarvar A. Kakhkhorov, Kyo Bin Kang, Nikolas Kessler, Roland D. Kersten, Hyunwoo Kim, Riley D. Kirk, Oliver Kohlbacher, Eftychia E. Kontou, Ken Liu, Itzel Lizama-Chamu, Gordon T. Luu, Tal Luzzatto Knaan, **Helena Mannochio-Russo**, Michael T. Marty, Yuki Matsuzawa, Andrew C. McAvoy, Laura-Isobel McCall, Osama G. Mohamed, Omri Nahor, Heiko Neuweger, Timo H. J. Niedermeyer, Kozo Nishida, Trent R. Northen, Kirsten E. Overdahl, Johannes Rainer, Raphael Reher, Elys Rodriguez, Timo T. Sachsenberg, Laura M. Sanchez, Robin Schmid, Cole Stevens, Shankar Subramaniam, Zhenyu Tian, Ashootosh Tripathi, Hiroshi Tsugawa, Justin J. J. van der Hooft, Andrea Vicini, Axel Walter, Tilmann Weber, Quanbo Xiong, Tao Xu, Tomáš Pluskal, Pieter C. Dorrestein, Mingxun Wang*
*Nature Methods*, Vol. 22, No. 6, 2025. doi.org/10.1038/s41592-025-02660-z

44. **Phytochemical Investigation and Biological Evaluation of *Disynaphia filifolia* : A Novel Source of Tremetone in Asteraceae**
Yasmin Rodrigues Chierici, Anderson Valdiney Gomes Ramos, Drielli Rhiane Peres Colhado Areas, Alana Carla Battistella, Solange Maria Cottica, Tatiana Shioji Tiuman, **Helena Mannochio-Russo**, Ana Lucia Tasca Gois Ruiz, Mary Ann Foglio, Marta Regina Barrotto do Carmo, Maria Helena Sarragiotto, Debora Cristina Baldoqui*
*Chemistry & Biodiversity*, Vol. 22, No. 9, 2025. doi.org/10.1002/cbdv.202402696

43. **The International Space Station has a unique and extreme microbial and chemical environment driven by use patterns**
Rodolfo A. Salido, Haoqi Nina Zhao, Daniel McDonald, **Helena Mannochio-Russo**, Simone Zuffa, Renee E. Oles, Allegra T. Aron, Yasin El Abiead, Sawyer Farmer, Antonio González, Cameron Martino, Ipsita Mohanty, Ceth W. Parker, Lucas Patel, Paulo Wender Portal Gomes, Robin Schmid, Tara Schwartz, Jennifer Zhu, Michael R. Barratt, Kathleen H. Rubins, Hiutung Chu, Fathi Karouia, Kasthuri Venkateswaran, Pieter C. Dorrestein*, Rob Knight*
*Cell*, Vol. 188, No. 7, 2025. doi.org/10.1016/j.cell.2025.01.039

42. **Discovery of metabolites prevails amid in-source fragmentation**
Yasin El Abiead, Adriano Rutz, Simone Zuffa, Bashar Amer, Shipei Xing, Corinna Brungs, Robin Schmid, Mario S. P. Correia, Andres Mauricio Caraballo-Rodriguez, Amir Zarrinpar, **Helena Mannochio-Russo**, Michael Witting, Ipsita Mohanty, Tomáš Pluskal, Wout Bittremieux, Rob Knight, Andrew D. Patterson, Justin J. J. van der Hooft, Sebastian Böcker, Warwick B. Dunn, Roger G. Linington, David S. Wishart, Jean-Luc Wolfender, Oliver Fiehn, Nicola Zamboni, Pieter C. Dorrestein*
*Nature Metabolism*, Vol. 7, No. 3, 2025. doi.org/10.1038/s42255-025-01239-4

41. **A guide to reverse metabolomics—a framework for big data discovery strategy**
Vincent Charron-Lamoureux, **Helena Mannochio-Russo**, Santosh Lamichhane, Shipei Xing, Abubaker Patan, Paulo Wender Portal Gomes, Prajit Rajkumar, Victoria Deleray, Andrés Mauricio Caraballo-Rodríguez, Kee Voon Chua, Lye Siang Lee, Zhao Liu, Jianhong Ching, Mingxun Wang, Pieter C. Dorrestein*
*Nature Protocols*, Vol. 20, No. 10, 2025. doi.org/10.1038/s41596-024-01136-2

40. **PERFIL DOS JOVENS DOUTORES EM QUÍMICA NO BRASIL: DESAFIOS, DISPARIDADES E PERSPECTIVAS**
**Helena Mannochio-Russo***, Fábio G. Delolo, Rayane C. F. Silva, Francisco L. F. da Silva, Eduarda A. Moreira, Vanessa Nascimento, Rodolfo G. Fiorot, Irlon M. Ferreira, Juliana C. Thomas, Wilhan D. G. Nunes, Alan C. Pilon, Romeu C. Rocha-Filho
*Química Nova*, 2025. doi.org/10.21577/0100-4042.20250148

39. **Chemical Characterization and Evaluation of the Anti-Cancer Potential of Flowers from Fridericia platyphylla (Bignoniaceae)**
Marcelino do Rosário, **Helena Mannochio-Russo**, Ana dos Santos, Aglaete Pinheiro, Luna Vasconcelos, Ana Paula Santos, Lila de Oliveira, Monique Martins, Marcelo de Andrade, Maria Nascimento, Vanderlan Bolzani, Josélia Lima, Cláudia da Rocha*
*Journal of the Brazilian Chemical Society*, 2025. doi.org/10.21577/0103-5053.20240132

38. **The changing metabolic landscape of bile acids – keys to metabolism and immune regulation**
Ipsita Mohanty, Celeste Allaband, **Helena Mannochio-Russo**, Yasin El Abiead, Lee R. Hagey, Rob Knight, Pieter C. Dorrestein*
*Nature Reviews Gastroenterology & Hepatology*, Vol. 21, No. 7, 2024. doi.org/10.1038/s41575-024-00914-3

37. **Special Issue in Honor of Professor Vanderlan da Silva Bolzani**
**Helena Mannochio-Russo**, Cristiano Soleo de Funari, Leticia V. Costa-Lotufo
*Journal of Natural Products*, Vol. 87, No. 3, 2024. doi.org/10.1021/acs.jnatprod.4c00195

36. **Co-occurrence network analysis reveals the alterations of the skin microbiome and metabolome in adults with mild to moderate atopic dermatitis**
Paulo Wender P. Gomes†, **Helena Mannochio-Russo**†, Junhong Mao, Haoqi Nina Zhao, Jacob Ancira, Craig D. Tipton, Pieter C. Dorrestein, Min Li*
*mSystems*, Vol. 9, No. 3, 2024. doi.org/10.1128/msystems.01119-23

35. **The underappreciated diversity of bile acid modifications**
Ipsita Mohanty†, **Helena Mannochio-Russo**†, Joshua V. Schweer, Yasin El Abiead, Wout Bittremieux, Shipei Xing, Robin Schmid, Simone Zuffa, Felipe Vasquez, Valentina B. Muti, Jasmine Zemlin, Omar E. Tovar-Herrera, Sarah Moraïs, Dhimant Desai, Shantu Amin, Imhoi Koo, Christoph W. Turck, Itzhak Mizrahi, Penny M. Kris-Etherton, Kristina S. Petersen, Jennifer A. Fleming, Tao Huan, Andrew D. Patterson, Dionicio Siegel, Lee R. Hagey, Mingxun Wang, Allegra T. Aron, Pieter C. Dorrestein*
*Cell*, Vol. 187, No. 7, 2024. doi.org/10.1016/j.cell.2024.02.019

34. **Effects of Crotalaria spectabilis extract in Meloidogyne javanica and Heterodera glycines soybean nematodes**
Tamires Scupinari, **Helena Mannochio-Russo**, Gustavo Galo Marcheafave, Vanderlan da Silva Bolzani, Estela de Oliveira Nunes, Waldir Pereira Dias, Clara Beatriz Hoffmann-Campo, Maria Luiza Zeraik*
*European Journal of Plant Pathology*, Vol. 170, No. 4, 2024. doi.org/10.1007/s10658-024-02821-4

33. **Prioritizing Mentorship as Scientific Leaders**
Jacky M. Deng*, Salma Elgaili Ahmed, Ernest Awoonor-Williams, Progna Banerjee, Magda H. Barecka, Laura E. Bickerton, Silvina A. Di Pietro, Stanna K. Dorn, Kevin Maik Jablonka, Gabriele Laudadio, Elisabeth Kreidt, **Helena Mannochio-Russo**, Júlio Terra, Olivia Harper Wilkins, Saigopalakrishna S. Yerneni, Maha Yusuf
*ACS Central Science*, Vol. 10, No. 2, 2024. doi.org/10.1021/acscentsci.3c00500

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    **Helena Mannochio-Russo**[*], Rafael F. de Almeida, Wilhan D. G. Nunes, Paula C. P. Bueno, Andrés M. Caraballo-Rodríguez, Anelize Bauermeister, Pieter C. Dorrestein, Vanderlan S. Bolzani[*]
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1. **Thermoanalytical and spectroscopic characteristics of young and old leaves powder and methanolic extracts of Niedenzuella multiglandulosa**
**Helena Mannochio Russo**†, Wilhan Donizete Gonçalves Nunes†, Vanderlan da Silva Bolzani, Massao Ionashiro, Flávio Junior Caires*
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## MANUSCRIPTS ACCEPTED FOR PUBLICATION

1. **plantMASST: a community-driven chemotaxonomic digitization of plants**

   **Helena Mannochio-Russo**[†], Xiaofeng Wang[†], Paulo Wender P. Gomes[†], Wilhan Donizete Gonçalves Nunes, Robin Schmid, Simone Zuffa, Luis Quiros-Guerrero, Tito Damiani, Wenjie Li, Derrick A. Ousley, Jordan A. Hungerford, Lisa S. Mydy, Clarissa Huisman, Matthew D. McConnachie, Hanyu Su, Gabrielle May, Yasin El Abiead, Andrés Mauricio Caraballo-Rodríguez, Haoqi Nina Zhao, Heejung Yang, Shipei Xing, Jehad Almaliti, Yahia Z. Tabaza, Jayden M. Roseberry, Deyvis-Dante Solis-Gozar, Harsha Gouda, Abubaker Patan, Julius Agongo, Vincent Charron-Lamoureux, Desnor Nicole Chigumba, Khadija Shafiq, Kali M. McDonough, Brian Edward Sedio, Jonathan A. Myers, Pierre-Marie Allard, Thomas V. Harwood, Giselle Tamayo-Castillo, Kyo Bin Kang, Emmanuel Defossez, Hector H.F. Koolen, Milton Nascimento da Silva, Consuelo Yumiko Yoshioka e Silva, Sergio Rasmann, Tom W. N. Walker, Gaëtan Glauser, José Miguel Chaves-Fallas, Bruno David, Hyunwoo Kim, Kyuhyeong Lee, Myeong Ji Kim, Won Jun Choi, Young-Sam Keum, Emilly J.S.P. de Lima, Lívia Soman de Medeiros, Giovana Anceski Bataglion, Emmanoel Vilaça Costa, Felipe Moura Araujo da Silva, Alice Rhelly V. Carvalho, José Diogo Evangelista Reis, Sônia das Graças Santa Rosa Pamplona, Eunah Jeong, Kyungha Lee, Geum Jin Kim, Yun-Seo Kil, Joo-Won Nam, Hyukjae Choi, Yoo Kyong Han, Si Young Park, Ki Yong Lee, Changling Hu, Yilun Dong, Shengmin Sang, Colin R. Morrison, Ricardo Moreira Borges, Andrew Magno Teixeira, Seo Yoon Lee, Bum Soo Lee, Se Yun Jeong, Ki Hyun Kim, Adriano Rutz, Arnaud Gaudry, Edouard Brülhart, Iris F. Kappers, Rumyana Karlova, Mara Meisenburg, Roland Berdaguer, J. Sebastián Tello, David Henderson, Rafael Felipe de Almeida, Anelize Bauermeister, Alan Cesar Pilon, Norberto Peporine Lopes, Luís Guilherme Pereira Feitosa, Marcelino Santos do Rosário, Kateřina Kučerová, Lúcia G. Lohmann, Vanderlan da Silva Bolzani, Paul E. Abraham, Leslie Eugenia Cayola, S. Joseph Wright, David N. Allen, Kristina J. Anderson-Teixeira, Jennifer L. Baltzer, James A. Lutz, Sean M. McMahon, Geoffrey Parker, John D. Parker, Trent R. Northen, Benjamin P. Bowen, Tomáš Pluskal, Justin J.J. van der Hooft, Jeremy J. Carver, Nuno Bandeira, Benjamin S. Pullman, Jing-Ke Weng, Jean-Luc Wolfender, Mingxun Wang, Pieter C. Dorrestein[*], Roland D. Kersten[*]
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## PREPRINTS

7. **An Annotation Consistency Checker for Evidence-Based Evaluation of Untargeted LC-MS/MS Metabolite Annotations**

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   **Helena Mannochio-Russo**[†], Patrick C. Ferreira[†], Kine Eide Kvitne, Abubaker Patan, Victoria Deleray, Julius Agongo, Harsha Gouda, Wilhan D. Gonçalves Nunes, Shipei Xing, Jasmine Zemlin, Martijn van Faassen, Erin R. Reilly, Imhoi Koo, Andrew D. Patterson, Shirley M. Tsunoda, Mingxun Wang, Dionicio Siegel, Lindsey A. Burnett, Pieter C. Dorrestein[*]
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1. **Charting the undiscovered metabolome with synthetic multiplexing implicates ibuprofen-carnitine in myotoxicity**
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## PRESENTATIONS

### Conferences

#### ORAL PRESENTATIONS

plantMASST: A community-driven mass spectrometry resource for the chemotaxonomic digitization of plants — 2026
74th ASMS Conference on Mass Spectrometry and Allied Topics, San Diego, CA, United States

The microbiome diversifies long- to short-chain fatty acid-derived N-acyl lipids — 2025
PACIFICHEM 2025, Honolulu, HI, United States

The microbiome diversifies N-acyl lipid pools - including short-chain fatty acid-derived compounds — 2025
Gordon Research Seminar - Metabolomics and Human Health, Ventura, CA, United States

The microbiome diversifies long- to short-chain fatty acid-derived N-acyl lipids 2025
2025 Metabolomics Association of North America (MANA) 7th Annual Conference, Banff, Alberta, Canada

The underappreciated diversity of bile acids modifications 2023
71st ASMS Conference on Mass Spectrometry and Allied Topics, Houston, TX, United States

Mass spectrometry-based untargeted metabolomics reveals unusual larvicidal aliphatic nitro esters from Heteropterys umbellata (Malpighiaceae) 2022
45a Reunião Anual da Sociedade Brasileira de Química, Maceió, AL, Brazil

Use of Statistical Tools and Automation to Accelerate the Natural Products Workflow – A Malpighiaceae Case Study 2021
II Bio.Natural-Bioactive Natural Products Research Meeting, Lisboa, Portugal (online)

Search for natural products of economic interest in fruit residues of Platonia insignis: bioactivity evaluation and structural characterization 2018
XXV Encontro de Química da Região Sul (SBQSul), Londrina, PR, Brazil

Chemical Profile of Tetrapterys multiglandulosa, a plant suspected of causing cattle intoxication in Brazil 2017
46th World Chemistry Congress, 40a Reunião Anual da Sociedade Brasileira de Química, São Paulo, SP, Brazil

### POSTER PRESENTATIONS

Pan-metabolomics repository mapping of the carnitine landscape 2026
74th ASMS Conference on Mass Spectrometry and Allied Topics, San Diego, CA, United States

The microbiome diversifies N-acyl lipid pools - including short-chain fatty acid-derived compounds 2025
Gordon Research Conference - Metabolomics and Human Health, Ventura, CA, United States

Community curation of microbiome metabolites for the mechanistic analysis of metabolomics data 2024
72nd ASMS Conference on Mass Spectrometry and Allied Topics, Anaheim, CA, United States

Repository scale discovery of microbial compounds include short chain fatty acid-derived metabolites 2024
20th Annual International Conference of the Metabolomics Society, Osaka, Japan

Mass spectrometry-based untargeted metabolomics reveals unusual larvicidal aliphatic nitro esters from Heteropterys umbellata (Malpighiaceae) 2022
45a Reunião Anual da Sociedade Brasileira de Química, Maceió, AL, Brazil

Chemotaxonomic investigations of Malpighiaceae botanical family using mass spectrometry-based metabolomics 2022
American Chemical Society Fall 2022 National Meeting, Chicago, IL, United States

Screening of Malpighiaceae Plant Extracts Against Aedes aegypti Larvae and Metabolomic Investigation of the Most Active Species 2021
8th Brazilian Conference on Natural Products, Goiania, GO, Brazil (online)

Untargeted metabolomics sheds light on the secondary metabolism of Malpighiaceae family 2021
69th International Congress and Annual Meeting of the Society for Medicinal Plant and Natural Product Research (GA), Bonn, Germany (online)

Experimental design applied in UPLC-DAD method development of Malpighiaceae species extracts and metabolite identification 2020
American Chemical Society Spring 2020 National Meeting, Philadelphia, PA, United States (online)

**Experimental planning for UPLC method development of hydroalcoholic extracts from Malpighiaceae species** 2019
XVII Latin American Symposium on Chromatography and Related Techniques, Aracaju, SE, Brazil

**Chemical profile of Crotalaria spectabilis extracts and evaluation of nematode control in Heterodera glycines** 2019
42a Reunião Anual da Sociedade Brasileira de Química, Joinville, SC, Brazil

**Identificação de metabólitos secundários com atividade anti-glicante em extratos da espécie Talisia esculenta** 2019
12o Simpósio Brasileiro de Farmacognosia, Petrópolis, RJ, Brazil

**Search for natural products of economic interest in fruit residues of Platonia insignis: bioactivity evaluation and structural characterization** 2018
XXV Encontro de Química da Região Sul (SBQSul), Londrina, PR, Brazil

**Chemical Profile of Tetrapterys multiglandulosa, a plant suspected of causing cattle intoxication in Brazil** 2017
46th World Chemistry Congress, 40a Reunião Anual da Sociedade Brasileira de Química, São Paulo, SP, Brazil

**Chemical profile and toxicity study of Tetrapterys multiglandulosa secondary metabolites using Zebrafish toxicity model** 2017
6th Brazilian Conference on Natural Products, Vitória, ES, Brazil

**Thermoanalytical characteristics of Tetrapterys multiglandulosa young and old leaves' powder and methanolic extracts by TG-DSC** 2017
VIII Simpósio de Análise Térmica, Ponta Grossa, PR, Brazil

### Other presentations

#### INVITED TALKS

**Unlocking the unexplored potential of metabolomics data: a journey from natural products to data science and clinical applications** 2025
San Diego State University Chemistry Department seminar, San Diego, CA, United States

**The microbiome diversifies long- to short-chain fatty acid-derived N-acyl lipids** 2025
UCSD HIV Neurobehavioral Research Program (HNRP) Depression Working Group Meeting, San Diego, CA, United States

## TEACHING / WORKSHOPS

#### TEACHING

**Extension course: “Basic concepts of mass spectrometry and introduction to the Global Natural Products Social Molecular Networking (GNPS) platform”** 2021
Faculty of Pharmaceutical Sciences, Federal University of Alfenas, MG, Brazil. December 6–10, 2021 (10 hours). 35 participants.

**Teaching assistant: “Experimental Organic Chemistry”** 2016
Institute of Chemistry, São Paulo State University. Feb/2016 to Jul/2016 (60 hours). 13 participants.

#### WORKSHOPS (ONLINE AND IN PERSON)

**Learning about the diversity of lipids from billions of MS/MS files at once - insights from molecular networking, query language and global fast search engines** 2025
Inserm Workshop 287 - Lipidomics: principles and applications in the field of health, Bordeaux, France. Estimated audience: 70 people.

**GNPS 2.0/CMMCkb Enrichment for Microbial Natural Product Discovery** 2025
International Conference on Natural Product Discovery and Development in the Genomic Era, San Diego, CA, United States. Estimated audience: 50 people.

**Classical Molecular Networking in GNPS2** 2024
Online. Estimated audience: 200 people.

**Public Data Re-use / Re-analysis with MetaboLights and GNPS** 2024
20th Annual International Conference of the Metabolomics Society, Osaka, Japan. Estimated audience: 50 people.

**Network Enrichment Analysis and Deposition Workflow – Collaborative Microbial Metabolite Center** 2023
Online. Estimated audience: 70 people.

**GNPS for Natural Product Discovery** 2023
International Conference on Natural Product Discovery and Development in the Genomic Era, San Diego, CA, United States. Estimated audience: 50 people.

**Molecular Networking with GC-MS data (in Portuguese)** 2021
Online. Estimated audience: 80 people.

**Feature-Based Molecular Networking (in Portuguese)** 2020
Online. Estimated audience: 200 people.

**Molecular Networking (in Portuguese)** 2020
Online. Estimated audience: 450 people.

## MENTORING

**Mentor to 4 undergraduate students and 2 graduate students** 2022 – 2026
University of California San Diego

**Mentor to 3 undergraduate students and 4 graduate students** 2018 – 2022
São Paulo State University

## ORGANIZATION OF EVENTS

**Understanding microbial metabolites and responses using LC-MS-based metabolomics** 2025
Two-day in-person workshop as part of the Collaborative Microbial Metabolite Center. San Diego, USA. Estimated audience: 75 people.

**Series of 4 seminars about GNPS2 in Portuguese** 2024
Online, as part of the Virtual Multi-Omics Lab (VMOL) seminars. Estimated audience: 80 to 200 people.

**Preparing proposals for research funding calls - what do funding agencies expect?** 2023
In-person workshop at the 46th Annual Meeting of the Brazilian Chemical Society. Águas de Lindoia, Brazil. Estimated audience: 50 people.

**Challenges for science in a continental country: a look at regional inequalities** 2023
In-person session at the 46th Annual Meeting of the Brazilian Chemical Society. Águas de Lindoia, Brazil. Estimated audience: 90 people.

**Young Researchers: future leaders in the academic and industrial sectors** 2019
In-person workshop at the 42nd Annual Meeting of the Brazilian Chemical Society. Joinville, Brazil. Estimated audience: 80 people.

## ADDITIONAL PROFESSIONAL ACTIVITIES

### PEER REVIEW

Reviewed 30+ manuscripts for the following journals: Applications in Plant Sciences, Bioorganic Chemistry, BMC Complementary and Alternative Medicine, Brazilian Journal of Pharmacology, Cell Host & Microbe, Communications Biology, Journal of Agricultural and Food Chemistry, Journal of Natural Products, Journal of the Brazilian Chemical Society, Nature Communications, Planta Medica, Scientific Data, Scientific Reports.

### LEADERSHIP / COMMUNITY ENGAGEMENT

**Co-chair of the Early Career Researchers Committee of the Brazilian Chemical Society** 2022 – 2024

As a co-chair of the Committee, I organized several workshops and round tables within the Annual Meeting of the Brazilian Chemical Society for promoting discussions of relevant topics for early career researchers in Brazil. One major project I coordinated was a national census of early career researchers, which gathered responses from over 400 participants through an online survey. The survey collected data on geographic distribution, career satisfaction, skills development, professional duties and rights, among others. The results were published in the Brazilian academic journal Química Nova.

**Member of the Early Career Researchers Committee of the Brazilian Chemical Society** 2019 – 2024

I served as a member of the Finances division (2019–2020) and the Workshops/Seminars division (2020–2022). Within the Finances division, I participated in a collaboration with the Royal Society of Chemistry to establish and sponsor the Early Career Researcher Award awarded yearly by the committee. As a member of the Workshops/Seminars division, I organized bi-monthly seminars on several topics relevant to early career researchers in Brazil.

### SCIENTIFIC SOCIETIES

**Member of the Brazilian Chemical Society** 2019 – Present

**Member of the Metabolomics Society** 2021 – Present

**Member of the American Society for Mass Spectrometry** 2022 – Present

**Member of the American Chemical Society** 2022 – Present

**Member of the Metabolomics Association of North America** 2024 – Present

## REFERENCES

Available upon request.

Last updated October 22, 2026